

# Lecture 11

Wednesday, August 13, 2025 10:06 AM



Lecture 11

## Engineering Mathematics III

ENGE702

Lecture 11: Second Order Nonhomogeneous ODEs

### Second order ODEs

From the table (in the last lecture), we know that our particular solution will be of the form

$$y_p = a \cos(\omega t) + b \sin(\omega t).$$

From here, we can take derivatives of  $y_p$ , and put it into our differential equation, and solve for  $a$  and  $b$ , and we get:

$$a = F_0 \frac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2} \quad \text{and} \quad b = F_0 \frac{\omega c}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$$

where

$$\omega_0 = \sqrt{\frac{k}{m}}$$

is called the 'natural frequency'.

By solving the 2nd order linear ODE, we obtain the particular solution  $y_p$ .

Understand the solution by looking at several cases.

Case 1. no damper,  $c=0$

$$\text{then } a = F_0 \frac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + 0} = \frac{F_0}{m(\omega_0^2 - \omega^2)}$$

$$b = 0$$

$$\therefore y_p = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

And the homogeneous soln from Lecture 8 is

$$y_h = A \sinh(\omega_0 t) + B \sin(\omega_0 t).$$

Hence:

$$y = y_h + y_p = A \sinh(\omega_0 t) + B \sin(\omega_0 t) + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

Case 2: Look at when  $\omega = \omega_0$ , the frequency of the external force is the same as the natural frequency  $\omega_0$ .

Then the particular solution has the form:

$$y_p = t(a \cos(\omega_0 t) + b \sin(\omega_0 t))$$

### Second order ODEs

#### Case 1: Undamped Forced Oscillations

Here  $c = 0$ . So in this case we have

$$a = \frac{F_0}{m(\omega_0^2 - \omega^2)} \quad \text{and} \quad b = 0$$

so our particular solution becomes

$$y_p = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

Our general solution is then

$$y = A \sin(\omega_0 t) + B \cos(\omega_0 t) + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

### Second order ODEs

We will now look at what happens when  $\omega = \omega_0$ . In this case, one of the terms in our homogeneous solution  $y_h$  contains a  $\cos(\omega_0 t)$  term, as does the suggested particular solution. So we need to use the modification rule, and our suggested particular solution becomes:

$$y_p = t(a \cos(\omega_0 t) + b \sin(\omega_0 t))$$

This gives (after finding the constants  $a$  and  $b$ ):

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$$y_p = \frac{F_0}{2m\omega_0} t \sin(\omega_0 t)$$

$$y_p = \boxed{t}(a \cos(\omega_0 t) + b \sin(\omega_0 t))$$

$\Rightarrow$  Solving for  $a$  and  $b$  yields

$$y_p = \frac{F_0}{2m\omega_0} t \sin(\omega_0 t)$$

## Second order ODEs

This gives displacements looking something like this:

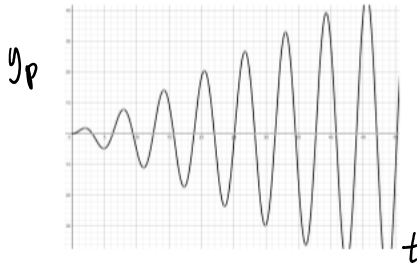


Figure: Resonance

In practical terms, this means the system would likely destroy itself, unless it caused it to be in a state where our model no longer applies.

## Second order ODEs

Another similar phenomenon is known as **beats**. Here the driving frequency is 'close' to the natural frequency.

We showed earlier that the general solution will be

$$y = A \sin(\omega_0 t) + B \cos(\omega_0 t) + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t) \rightarrow \text{could be quite large.}$$

Suppose we have a particular solution (for a particular set of initial conditions) that is

$$y = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t) - \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega_0 t)$$

$\omega_0^2 - \omega^2$  is small.

## Second order ODEs

This can be rewritten as:

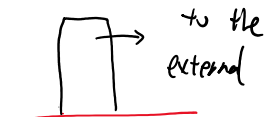
$$y = \frac{2F_0}{m(\omega_0^2 - \omega^2)} \sin\left(\frac{\omega_0 + \omega}{2} t\right) \sin\left(\frac{\omega_0 - \omega}{2} t\right)$$

When  $\omega_0$  and  $\omega$  are close, then the period of the last sin function is large. So we end up with something like:



Large displacements

building  $\omega_0$  different



frequency  $\omega$   
to avoid collapsing.

## Second order ODEs

### Case 2: Damped Forced Oscillations

Again the solution is a combination of the homogeneous solution and a particular solution.

But recall from the section on unforced oscillation that the unforced damped system will always tend to 0 as time increases. This means that:

'After a sufficiently long time the output of a damped vibrating system under a purely sinusoidal driving force will practically be a harmonic oscillation whose frequency is that of the input.'

So for damped forced oscillations, there is the **transient phase**, where the homogeneous solution will be involved, ie  $y = y_h + y_p$ , and the **steady-state phase**, where the homogeneous solution is negligible, so  $y = y_p$ .

For the damped forced oscillation:

$$y = y_h + y_p$$

we know that  $y_h \rightarrow 0$  eventually,

so the displacement will approximately equal to  $y_p$   $\lim_{t \rightarrow \infty} y(t) = \lim_{t \rightarrow \infty} y_h + y_p \approx y_p(t)$

Rewrite the particular solution:

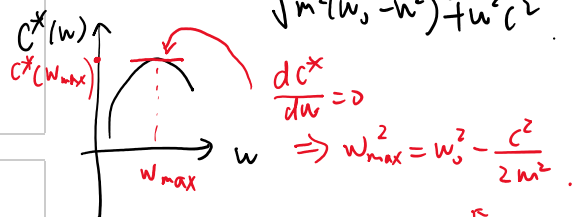
$$y_p = a \cos(\omega t) + b \sin(\omega t)$$

$$= C^* \cos(\omega t - \eta)$$

Amplitude phase angle.

The amplitude can be expressed as a function of the driving frequency:

$$C^*(\omega) = \frac{F_0}{\sqrt{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}}$$



$$C^*(\omega_{max}) = \frac{2mF_0}{c\sqrt{4m^2\omega_0^2 - c^2}}$$

The largest amplitude in a damped oscillation.

## Second order ODEs

In the damped case, we do not have 'run-away' resonance. Oscillations will always be finite (even theoretically), but some form of resonance still exists. This is called **practical resonance**.

The first thing we will note is that  $y_p(t) = a \cos(\omega t) + b \sin(\omega t)$  can be rewritten

$$y_p = C^* \cos(\omega t - \eta)$$

where  $C^*$  is called the **amplitude** and  $\eta$  is called the **phase angle**.

We can derive (see the textbook for details) that the **steady state amplitude** at a particular frequency  $\omega$  is given by:

$$C^*(\omega) = \frac{F_0}{\sqrt{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}}$$

## Second order ODEs

After some further derivation, we can find that the frequency at which the greatest amplitude occurs, called  $\omega_{max}$ , is given by

$$\omega_{max}^2 = \omega_0^2 - \frac{c^2}{2m^2}$$

with a corresponding amplitude of

$$C^*(\omega_{max}) = \frac{2mF_0}{c\sqrt{4m^2\omega_0^2 - c^2}}$$

## Second order ODEs

**Example:** Find the frequency  $\omega$  that would give the greatest amplitude of oscillation in a system modelled with the following differential equation

$$2y'' + 4y' + 7y = 4 \cos(\omega t)$$

Example: For  $2y'' + 4y' + 7y = 4 \cos(\omega t)$ .

$$m=2, \quad c=4, \quad k=7, \quad F_0=4.$$

$$\omega_{max}^2 = \omega_0^2 - \frac{c^2}{2m^2} = \frac{k}{m} - \frac{c^2}{2m^2} = \frac{7}{2} - \frac{16}{2 \times 2^2} = \frac{3}{2}$$

$$\Rightarrow \omega_{max} = \sqrt{\frac{3}{2}}$$

$$\text{and } C^*(\omega_{max}) = \frac{2mF_0}{c\sqrt{4m^2\omega_0^2 - c^2}} = \frac{2 \times 2 \times 4}{4\sqrt{4 \times 2^2 \times \frac{7}{2} - 4^2}} = \frac{1}{\sqrt{\frac{5}{2}}} = \frac{\sqrt{2}}{\sqrt{5}}$$

## Second order ODEs

In this lecture we have so far dealt with a mechanical forced mass - spring - damper system. There is a completely analogous electrical system: that of a resistor - inductor - capacitor (RLC) circuit with an applied voltage.

If we have a resistance  $R$ , an inductance  $L$ , a capacitance  $C$ , and an EMF of  $E_0 \sin(\omega t)$ , the current in the circuit,  $I$ , is modelled by

$$LI'' + RI' + \frac{1}{C}I = E_0 \omega \cos(\omega t)$$

We will not go into the details of this in this lecture (it is covered in Chapter 2.9 in the textbook), but will note that there is a direct analogy between this model and the mechanical model we have studied today.

## Exercises

- ▶ Find the steady state amplitude and the frequency of the systems modelled by the following differential equations:
  - ▶  $3y'' + 2y' + y = 5 \cos(4t)$
  - ▶  $y'' + 3y' + 6y = 2 \cos(13t)$
- ▶ What driving frequency will give the maximum amplitude of oscillation for each of the above systems, and what will be the amplitude at this frequency?

## Text Book References

Chapter 2.8